Computer Science practice paper

OCR GCSE Computer Science (J277), Paper 1 topics: Computer systems

Time allowed: 30 minutes **Total: 30 marks** **Topics: 1.1 to 1.6**

Name: _____

Instructions

- Answer all the questions in the spaces provided. Use black ink.
- You may not use a calculator, as in the real Paper 1. Show your working.
- The marks for each question are shown in brackets: roughly one mark a minute.
- This is a practice paper written in the style of the real exam. It is not an OCR paper.

Question 1: Systems architecture

1(a) State the purpose of the CPU. **[1 mark]**

1(b) Which register holds the address of the next instruction to be fetched? Choose one. **[1 mark]**

- [ ] Accumulator
- [ ] Memory Address Register (MAR)
- [ ] Memory Data Register (MDR)
- [ ] Program Counter (PC)

1(c) Explain how increasing the number of cores in a CPU can improve its performance. **[2 marks]**

1(d) Give one example of an embedded system. **[1 mark]**

Question 2: Memory and storage

2(a) Explain two differences between RAM and ROM. **[2 marks]**

2(b) Convert the binary number 10110110 into denary. **[1 mark]**

2(c) Convert the denary number 75 into an 8-bit binary number. **[1 mark]**

2(d) Convert the hexadecimal number 3F into denary. **[1 mark]**

2(e) Add the binary numbers 01101100 and 00110101. Give your answer as an 8-bit binary number. Show your working. **[2 marks]**

2(f) An image is 100 pixels wide and 50 pixels high. It has a colour depth of 8 bits. Calculate the file size of the image in bytes. Ignore metadata. Show your working. **[2 marks]**

2(g) Describe one difference between lossy and lossless compression. **[2 marks]**

Question 3: Networks

3(a) State what is meant by a LAN. **[1 mark]**

3(b) A school uses a star topology. Explain one advantage of a star topology. **[2 marks]**

3(c) Which protocol is used to send emails? Choose one. **[1 mark]**

- [ ] FTP
- [ ] HTTP
- [ ] IMAP
- [ ] SMTP

3(d) Describe what a DNS (Domain Name Server) does when a user types a web address into a browser. **[2 marks]**

Question 4: Network security and systems software

4(a) Describe what is meant by phishing. **[2 marks]**

4(b) Explain how a firewall helps to protect a network. **[2 marks]**

4(c) Identify one function of an operating system and describe what it does. **[2 marks]**

Question 5: Legal issues

5 State two principles of the Data Protection Act 2018 that an online shop must follow with its customers' data. **[2 marks]**

End of questions. Check your answers if you have time left.